WASP-10b

R-1600

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_210903` · created 2026-08-02T02:14:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459460.859 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.015
Transit depth (Rp/Rs)^2	3.04 +/- 0.51 [%]
Orbital Inclination (inc)	89.97 +/- 0.68
Airmass coefficient 1 (a1)	1.058 +/- 0.02
Airmass coefficient 2 (a2)	-0.052 +/- 0.043
Scatter in the residuals of the lightcurve fit is	1.9 %
Best Comparison Star	#2 - [617, 172]
Optimal Aperture	3.27
Optimal Annulus	15.67
Transit Duration (day)	0.0959 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.015 vs 0.1592 (deviation 0.69σ)
Duration fit vs published	0.0959 d vs 0.0946 d (deviation 0.61σ)
Mid-time O–C	-0.3 min
Depth SNR	8.1
Residual scatter	1.9 %

Light curve

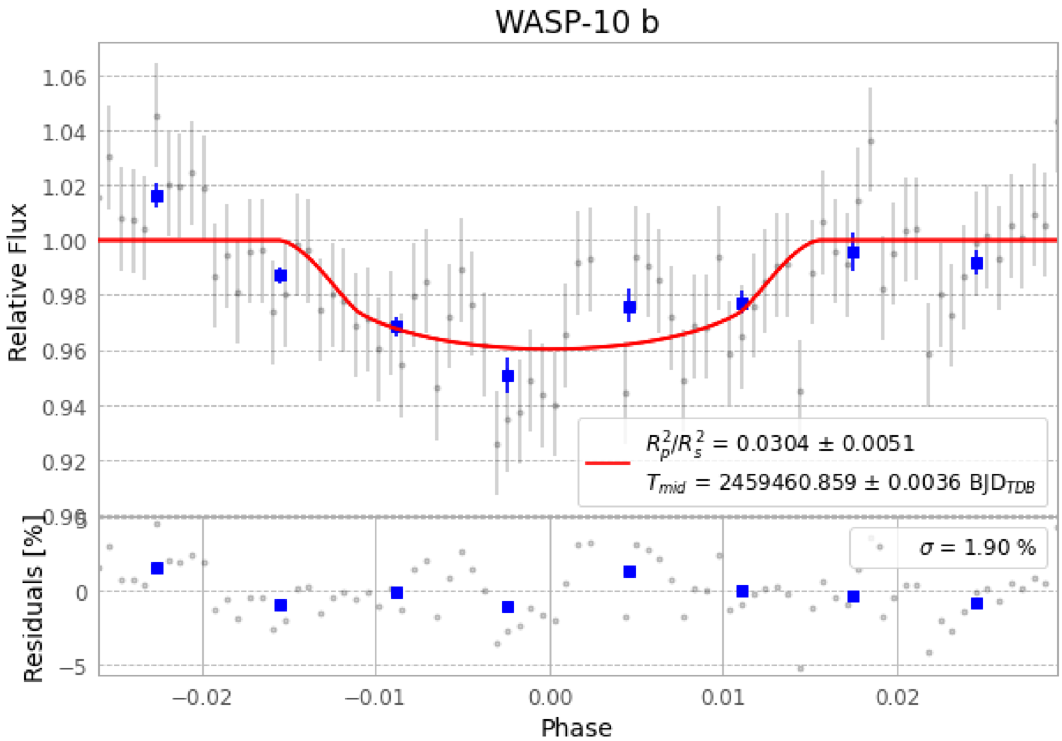

Figure 1 — FinalLightCurve_WASP-10 b_2021-09-02.png (SHA-256 0ce3ebd6a8f07533…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 162], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9bbb70494e2d7aba…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

83 FITS frames (55 MB), `WASP-10210903063312.FITS` → `WASP-10210903103912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-210903.log (7e4351becea5…), FinalLightCurve_WASP-10 b_2021-09-02.png (0ce3ebd6a8f0…), FinalLightCurve_WASP-10 b_2021-09-02.csv (2734aae872e3…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)